EXAMINATION QUESTION DIFFICULTY ANALYSIS REPORT

Python Project Report

1.Overview

The Examination Question Difficulty Analysis Report is a Python-based project designed to analyze examination questions and classify them according to their difficulty levels. The main objective of this project is to identify whether a question is Easy, Medium, or Hard based on specific criteria such as question length, complexity, keywords, and marks assigned.

This project uses Python programming concepts, including string processing, conditional statements, functions, and data analysis techniques, to evaluate questions and generate a difficulty analysis report. It helps teachers and educators understand the distribution of questions across different difficulty levels and assess whether an examination paper is properly balanced.

The system can process a set of examination questions, analyze their characteristics, categorize them into appropriate difficulty levels, and display a summary of the results. The generated report can help educators improve question paper design and maintain a suitable balance between simple, moderate, and challenging questions.

Overall, this project demonstrates how Python can be used to automate question analysis and support the preparation of balanced and effective examination papers.

2.Example Record Format

Question ID	Subject	Question	Marks	Difficulty
Q001	Python	What is Python?	2	Easy
Q002	Python	Explain Python data types.	5	Medium
Q003	Python	Explain OOP concepts with examples.	10	Hard

Difficulty Levels: Easy, Medium, Hard.

3.Problem Statement

The Examination Question Difficulty Analysis Report aims to analyze examination questions and classify them into Easy, Medium, and Hard levels based on question complexity, keywords, and marks. This system helps teachers evaluate the difficulty distribution of questions and prepare balanced question papers.

4.Objectives

1. To analyze examination questions using Python.

2. To classify questions into Easy, Medium, and Hard levels.
3. To calculate the percentage of questions in each difficulty level.
4. To generate a summary report of question difficulty.
5. To help teachers prepare balanced examination papers.

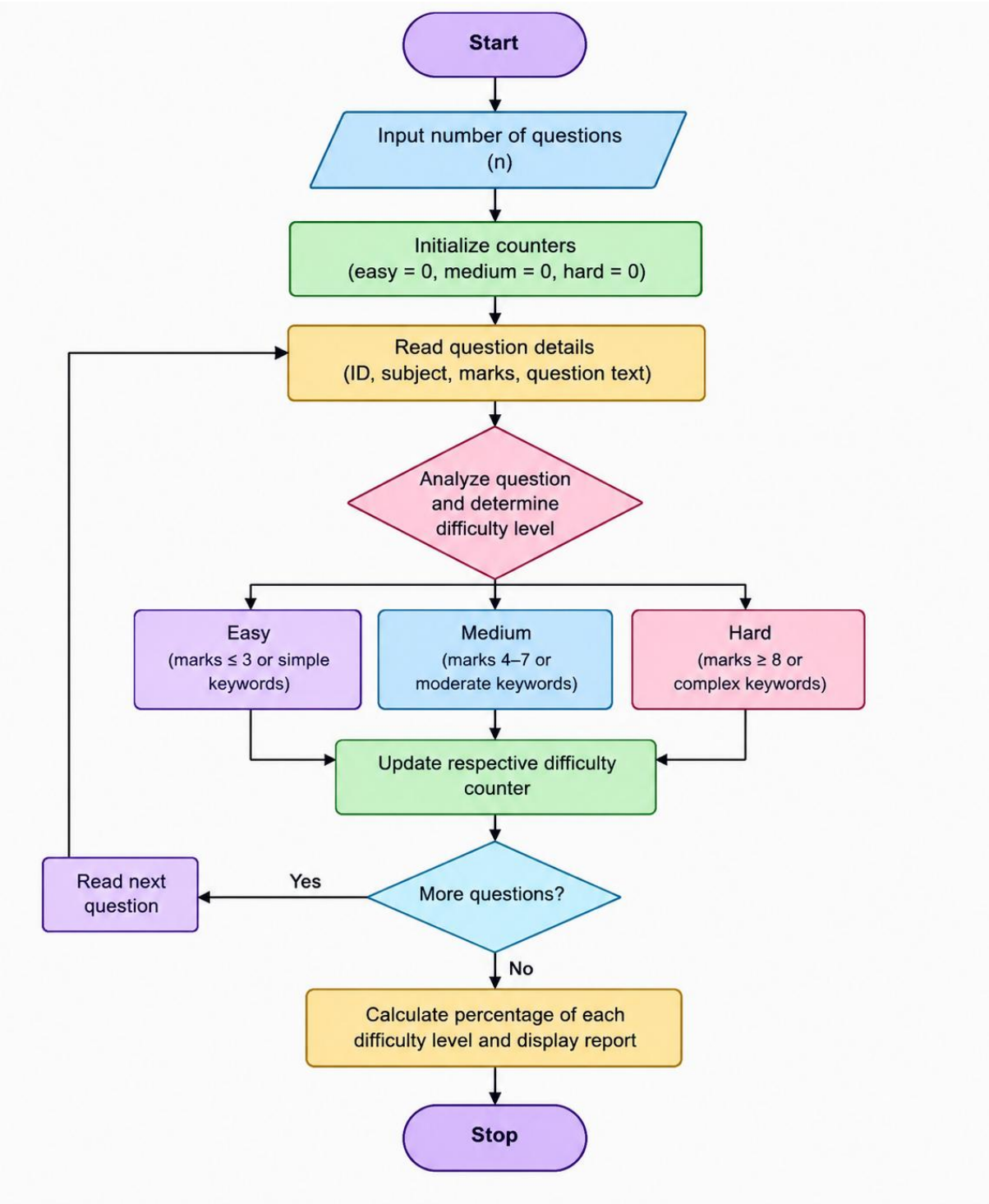
5.Assumptions

1. Examination questions are provided as input.
2. Each question has a unique question ID.
3. Marks and question text are available for analysis.
4. Difficulty levels are assigned using predefined rules.
5. The input data is assumed to be valid and complete.

6.Algorithm

1. Start the program.
2. Enter the examination questions and their marks.
3. Read each question.
4. Analyze the question using predefined difficulty rules.
5. Classify each question as Easy, Medium, or Hard.
6. Count the questions in each category.
7. Calculate the percentage of each difficulty level.
8. Display the analysis report.
9. Stop the program.

7.Flowchart



8.Pseudocode

BEGIN

INPUT number of questions

SET Easy = 0, Medium = 0, Hard = 0

```
  FOR each question
    INPUT question details and marks

    IF marks <= 3
      Easy = Easy + 1
    ELSE IF marks <= 7
      Medium = Medium + 1
    ELSE
      Hard = Hard + 1
    END IF
  END FOR

  DISPLAY Easy, Medium, Hard counts
  CALCULATE difficulty percentages
  DISPLAY analysis report
END
```

9.Source Code

```
# Examination Question Difficulty Analysis Report

def analyze_questions():
  print("=" * 45)
  print(" EXAMINATION QUESTION DIFFICULTY ANALYSIS")
  print("=" * 45)

  try:
    n = int(input("Enter number of questions: "))

    if n <= 0:
```

```
    print("Please enter a positive number.")
    return

easy = 0
medium = 0
hard = 0

for i in range(1, n + 1):
    print("\nQuestion", i)

    question = input("Enter question text: ")
    marks = int(input("Enter marks: "))

    if marks <= 0:
        print("Marks must be greater than zero.")
        return

    if marks <= 3:
        level = "Easy"
        easy += 1
    elif marks <= 7:
        level = "Medium"
        medium += 1
    else:
        level = "Hard"
        hard += 1

    print("Question:", question)
```

```
            print("Difficulty Level:", level)

        print("\n" + "=" * 45)
        print("         ANALYSIS REPORT")
        print("=" * 45)

        print("Total Questions:", n)
        print("Easy Questions:", easy)
        print("Medium Questions:", medium)
        print("Hard Questions:", hard)

        print("\nDifficulty Percentages:")
        print("Easy: {:.2f}%".format(easy / n * 100))
        print("Medium: {:.2f}%".format(medium / n * 100))
        print("Hard: {:.2f}%".format(hard / n * 100))

        print("=" * 45)
        print("Analysis Completed Successfully!")

    except ValueError:
        print("Invalid input! Please enter numbers for question count and marks.")


if __name__ == "__main__":
    analyze_questions()
```

10.Test Cases

Test Case ID	Input	Expected Output

TC01	Marks = 2	Difficulty: Easy
TC02	Marks = 5	Difficulty: Medium
TC03	Marks = 10	Difficulty: Hard
TC04	Number of Questions = 3	Accepts 3 questions
TC05	Number of Questions = 0	Displays an error message
TC06	Invalid marks input (text)	Displays an invalid input message
TC07	Marks = -2	Displays an error message
TC08	3 questions with marks 2, 5, 10	Displays counts and percentages correctly

TEST RESULT

The test cases are designed to verify question classification, input validation, and percentage calculation. Actual results should be checked by running the program.

11.Output Screenshots

```
==============================================
 EXAMINATION QUESTION DIFFICULTY ANALYSIS
==============================================
Enter number of questions: 3

Question 1
Enter question text: what is python?
Enter marks: 2
Question: what is python?
Difficulty Level: Easy

Question 2
Enter question text: Explain Python Data Types?
Enter marks: 5
Question: Explain Python Data Types?
Difficulty Level: Medium

Question 3
Enter question text: Explain 00P concepts.
Enter marks: 10
Question: Explain 00P concepts.
Difficulty Level: Hard

==============================================
           ANALYSIS REPORT
==============================================
Total Questions: 3
Easy Questions: 1
Medium Questions: 1
Hard Questions: 1

Difficulty Percentages:
Easy: 33.33%
Medium: 33.33%
Hard: 33.33%
==============================================
Analysis Completed Successfully!
```

12.Conclusion

The Examination Question Difficulty Analysis Report is a Python-based project that classifies examination questions into Easy, Medium, and Hard levels based on marks. It calculates the number and percentage of questions in each category and displays a summary report. This project helps teachers analyze question paper difficulty and prepare balanced examinations. It demonstrates the practical use of Python concepts such as functions, loops, conditional statements, and input validation.